

Masterclass on nonnegative Low-Rank Approximations

Lecture 2 - Nonnegative regression: NNLS, NNKL, and algorithms

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Nonnegative regression

In alternating optimization,

$$\min_{W \geq 0, H \geq 0} \|Y - WH^T\|_F^2$$

fixing all but one factor turns a nonnegative LRA problem into a nonnegative regression problem.

$$\min_{W \geq 0} \|Y - WH^T\|_F^2$$

Today: understand this optimization problem and the algorithms that solve it.

From a matrix to a vector problem

The problem is separable rowwise,

$$\|Y - WH^T\|_F^2 = \sum_{i=1}^m \|Y[i, :] - W[i, :]H^T\|_2^2.$$

We need to minimize a quadratic form for each rows $W[i, :] := x$, leading to solving m problems of the form

$$\min_{x \geq 0} \|y - Ux\|_2^2.$$

Nonnegative regression

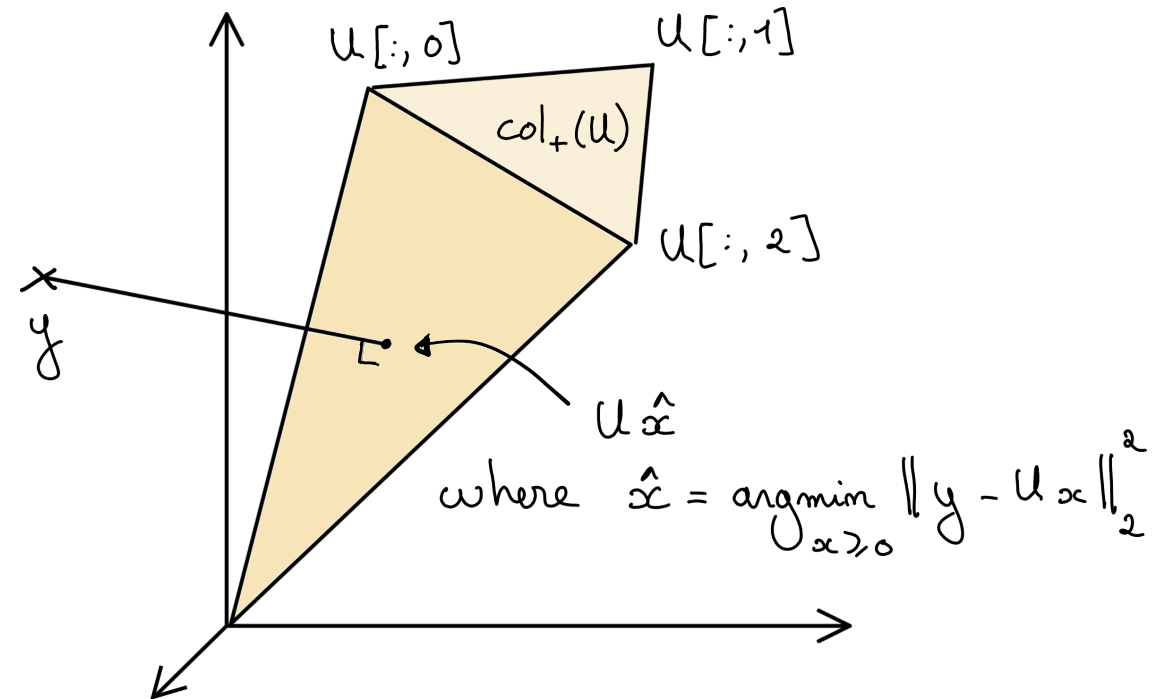
Given observations $y \in \mathbb{R}^m$ and a dictionary $U \in \mathbb{R}^{m \times n}$:

$$\min_{x \geq 0} f(y, Ux).$$

- x : nonnegative coefficients
- Ux : reconstructed observation
- f : separable, convex loss in its second argument

Two losses will organize the lecture:

- Euclidean distance: **NNLS**
- Kullback-Leibler divergence: **NNKL**



Why nonnegativity matters

Interpretation

Without constraints, least squares gives

$$x_{\text{LS}} = U^\dagger y.$$

Negative coefficients can be difficult to interpret as abundances, intensities...

- Question: is this clear to everyone ?

Solution uniqueness and properties

Nonnegativity is an informative prior:

- underdetermined problems can still have unique solutions
- solutions are naturally sparse
- feasible reconstructions form a cone

NNLS: the Euclidean case

Problem formulation

The nonnegative least-squares problem is

$$\text{NNLS: } x^* \in \arg \min_{x \geq 0} \|y - Ux\|_2^2$$

- It is a convex quadratic program.
- A solution always exists (projection on a convex set, see next slide)
- There is generally no closed-form formula.

For full columns rank matrix U , the projected estimate $[U^\dagger y]^+$ is generally **not** the NNLS solution.

NNLS as projection on a cone

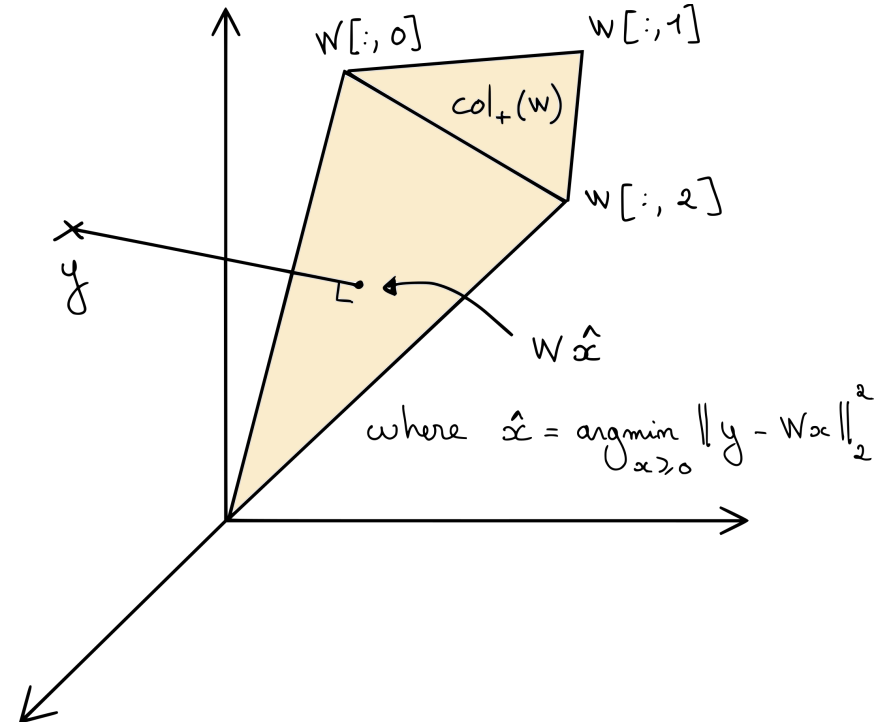
Define the cone generated by the columns of U :

$$\mathcal{C}(U) = \{Ux : x \geq 0\}.$$

Then NNLS can be written as

$$\min_{z \in \mathcal{C}(U)} \|y - z\|_2^2.$$

The reconstructed vector $z^* = Ux^*$ is the orthogonal projection of y onto the cone.



When y lies outside the cone, its projection often lands on a lower-dimensional facet. The corresponding coefficient vector x^* is sparse.

Caution

The actual shape of the cone depends on matrix U . Most of the intuition in this masterclass assumes that U is nonnegative.

KKT conditions

The geometry gives intuition. The KKT conditions transform this intuition into a set of equations.

Consider the Lagrangian

$$\mathcal{L}(x, \lambda) = \|y - Ux\|_2^2 - \langle \lambda, x \rangle,$$

an optimal pair (x^*, λ^*) satisfies

$$\begin{aligned} x^* &\geq 0, & \lambda^* &\geq 0, \\ \lambda^*[i]x^*[i] &= 0, \\ 2U^T(Ux^* - y) - \lambda^* &= 0. \end{aligned}$$

These conditions separate active coefficients from inactive ones.

KKT conditions on the support

Let $S = \text{supp}(x^*)$ and $r^* = y - Ux^*$.

On the support:

$$U[:, S]^T r^* = 0.$$

Outside the support:

$$U[:, S^c]^T r^* \leq 0.$$

So the active atoms are orthogonal to the residual.

Tip

If the optimal support is known and $U[:, S]$ is invertible, the solution can be computed exactly!

$$U^T[:, S]U[:, S]x^* = U^T[:, S]y$$

The night-sky theorem

Theorem

If $y \notin \mathcal{C}(U)$ and $\text{spark}(U) > m$, then NNLS has a unique solution with

$$|\text{supp}(x^*)| \leq m - 1.$$

This is the **night-sky theorem**:

- an underdetermined NNLS problem can have a unique solution.
- the solution is sparse even without an explicit sparsity penalty.
- knowing the support reduces the problem to least squares.

Proof elements

- The residual cannot be null, and is uniquely defined.
- The condition $U[:, S]^T r^* = 0$ imposes that columns of $U[:, S]$ are in the orthogonal space of r^* , of dimension $m - 1$.
 - Thus $U[:, S]$ is also always full column rank, and there exists a solution with a support of size at most $m - 1$.
- If there are no groups of m colinear columns (the spark condition), then S is unique, and so is x^* .

NNLS recap

NNLS is simultaneously:

- a constrained quadratic program.
- a projection onto a convex cone.
- a support-estimation problem.

Its algorithms will exploit one of these three viewpoints. We will explore them in a second part of the talk.

NNKL: the Poisson case

Why replace the Euclidean loss?

The Euclidean loss treats absolute errors equally across the signal range.

For data with large dynamic range or count measurements:

- small values may be scientifically important
- variance can grow with the signal
- relative error can matter more than absolute error

$$(100 - 101)^2 = 1$$

$$(1 - 2)^2 = 1$$

A Poisson model gives a different data fidelity term.

NNKL problem formulation

For Poisson observations (distribution of counting things),

$$Y[i] \mid x \sim \mathcal{P}((Ux)[i]),$$

maximum likelihood leads to the generalized KL divergence

$$D_{\text{KL}}(y, z) = \sum_i y[i] \log \frac{y[i]}{z[i]} + z[i] - y[i].$$

The nonnegative KL regression problem is

$$\text{NNKL: } x^* \in \arg \min_{x \geq 0} D_{\text{KL}}(y, Ux).$$

KL divergence in one dimension

The scalar KL divergence is minimized at $q = p$.

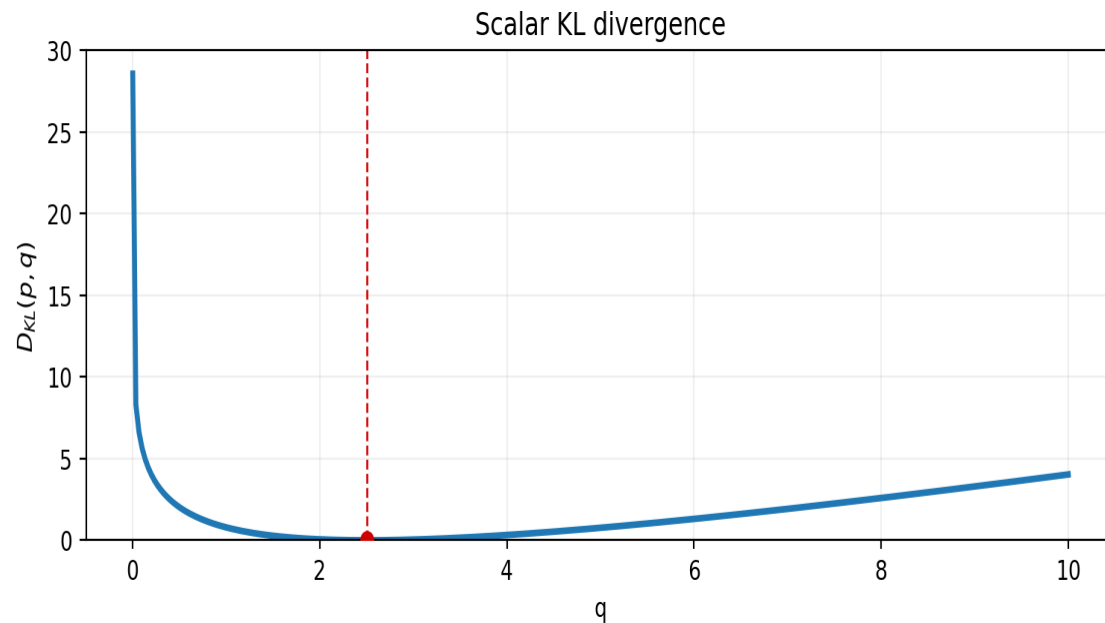
$$D_{\text{KL}}(p, q) = p \log\left(\frac{p}{q}\right) + q - p.$$

For the reference value $p = 2.5$, the curve is:

$$D_{\text{KL}}(100, 101) = 100 \log(100/101) + 1 \approx 0.005$$

$$D_{\text{KL}}(1, 2) = \log(1/2) + 1 \approx 0.31$$

$$D_{\text{KL}}(2, 1) = 2 \log(2) - 1 \approx 0.39$$



NNLS and NNKL: different geometries

NNLS

$$\|y - Ux\|_2^2$$

- quadratic
- orthogonal projection
- closed form solution if the support is known

NNKL

$$D_{\text{KL}}(y, Ux)$$

- Poisson likelihood
- non-Euclidean projection
- no closed-form even with support (see later)

Difficulty 1: curvature near zero

The KL curvature is

$$\frac{\partial^2 D_{\text{KL}}(y, z)}{\partial z^2} = \frac{y}{z^2}.$$

It becomes unbounded as $z \rightarrow 0$.

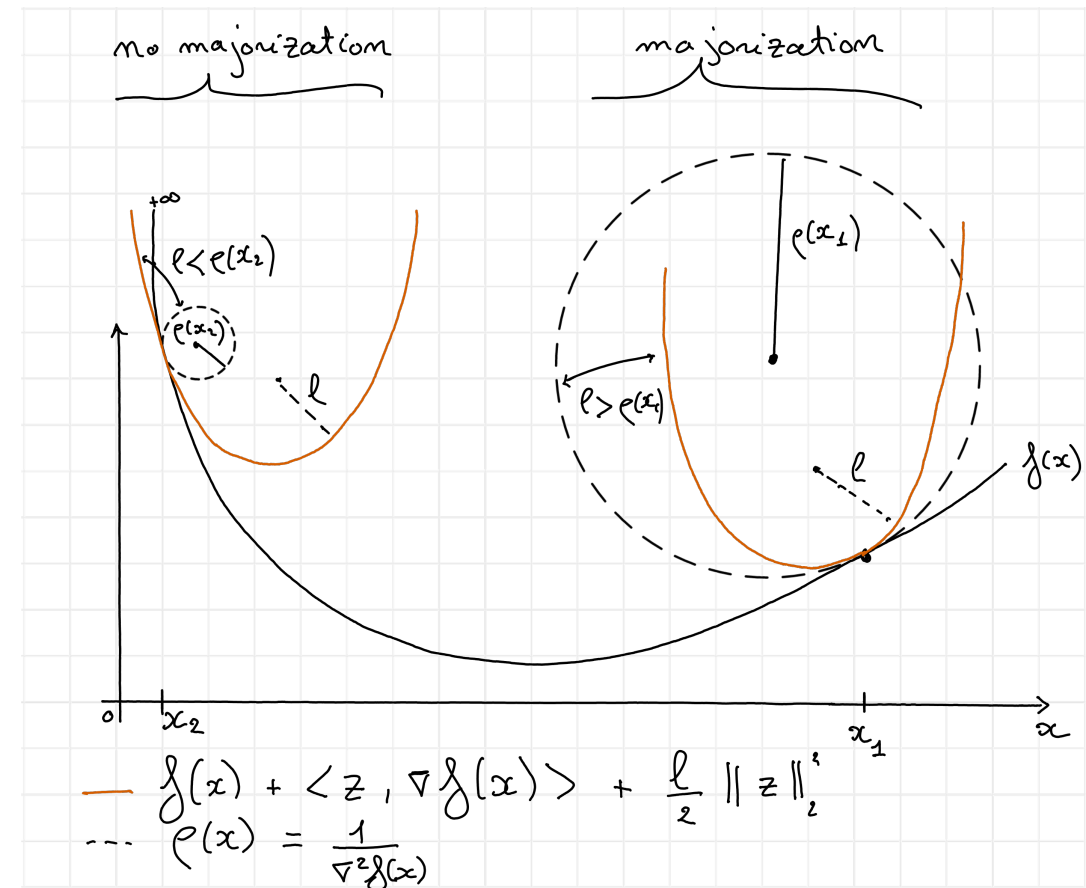
Consequences:

- No global Lipschitz constant for the gradient: $\forall y \geq 0, \forall L \geq 0, \exists z \geq 0$ s.t. $\|\nabla_x D_{\text{KL}}(y, z)\| > L\|z\|$.
- No universally safe fixed gradient step.
- Sparse data can make first-order methods inefficient.

Difficulty 2: no strong convexity

KL divergence is strictly convex, but generally not strongly convex: for large z , it becomes asymptotically linear.

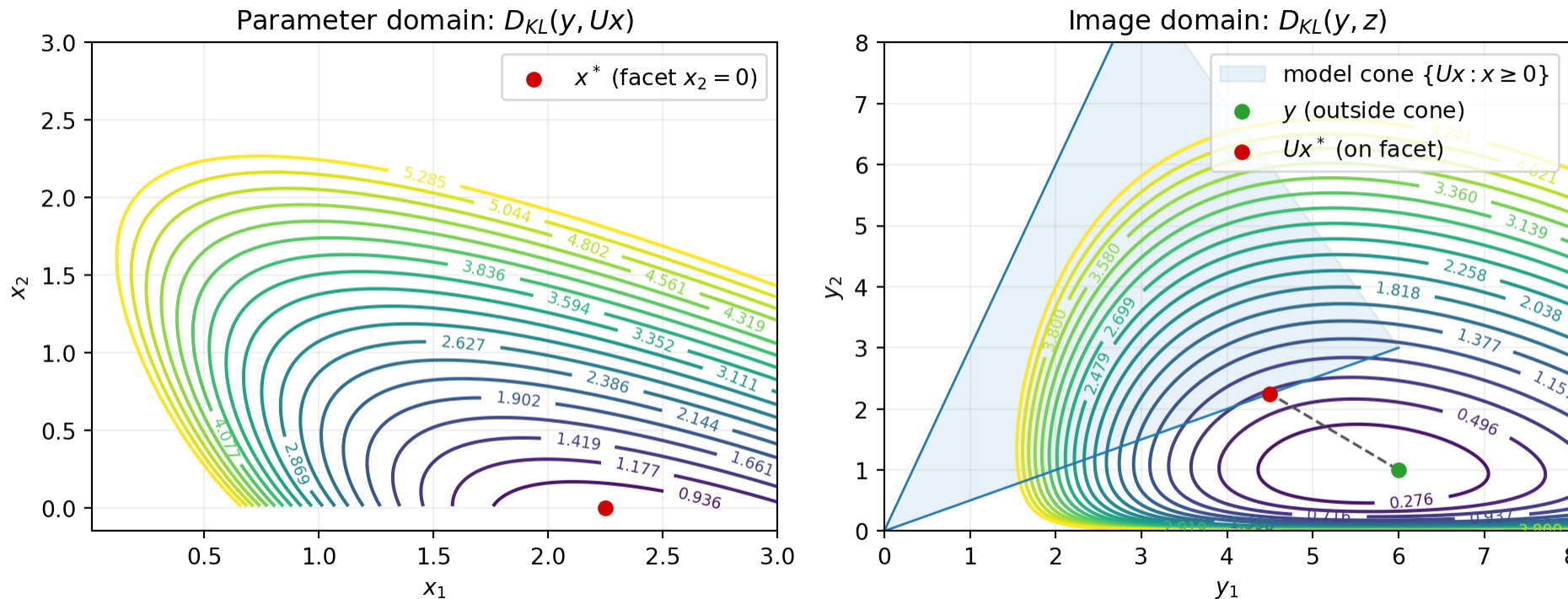
- gradient methods need not have a global linear rate
- large dynamic ranges make the problem harder
- zeros in y create broad linear regions



Difficulty 3: the support is not enough

On a known support S , the NNKL KKT equation is

$$U[:, S]^T \left(1 - \frac{y}{Ux^*}\right) = 0.$$



$y = [6. \ 1. \ 2.]$, $Ux^* = [4.499999918 \ 2.250000209 \ 2.250000009]$, residual $1-y/(Ux^*) = [-0.333333358 \ 0.555555597 \ 0.111111115]$

Here y lies outside the model cone $\{Ux : x \geq 0\}$: solving the KKT equation on the full support $\{1, 2\}$ would give a solution violating nonnegativity, so the true projection Ux^* falls on a facet of the cone ($x_2 = 0$), leaving column 2 inactive. Even locating the solution on this facet requires solving an optimization problem.

Bonus trick: optimal scaling for NNKL

Consider scaling a candidate reconstruction by α :

$$\min_{\alpha \geq 0} D_{\text{KL}}(y, \alpha Ux).$$

At the optimum,

$$\alpha^* = \frac{\sum_i y[i]}{\sum_i (Ux)[i]}.$$

Thus any optimal solution satisfies

$$\sum_i (Ux^*)[i] = \sum_i y[i].$$

When columns of U sum to one, the model and data have matching marginals.

Three algorithmic viewpoints

Which algorithm for which problem?

Method	NNLS	NNKL	Main idea
Active-Set	yes	no	search for the support
HALS	yes	no	exact coordinate blocks
Multiplicative Updates	yes	yes	scaled first-order steps

The best method depends on the shape of the regression:

- one vector or a matrix of right-hand sides?
- Euclidean or KL loss?
- sparse solution or dense iterates?

Active-Set for NNLS

Active-Set: search for the support

If the support S were known, NNLS would reduce to

$$\min_{x[S]} \|y - U[:, S]x[S]\|_2^2.$$

Active-Set (AS) alternates between:

1. adding a promising atom
2. solving least squares on the active support
3. removing coefficients that violate nonnegativity

Step 1: select an atom

With residual $r = y - Ux$, choose the inactive atom most correlated with r :

$$j = \arg \max_{i \notin S} \langle U[:, i], r \rangle, \quad S \leftarrow S \cup \{j\}.$$

- This follows the KKT condition: at optimality, correlations vanish on the support and are negative outside it.
- Other perspective: choose the column of U that is most correlated with the residual, likely to “explain” the remaining data the best.

Step 2: solve on the active support

Compute the unconstrained restricted least-squares solution

$$z[S] = U[:, S]^{\dagger} y, \quad z[S^c] = 0.$$

- If $z \geq 0$, accept it and continue with the next iteration. Stop if KKT conditions show optimality.
- If some entries of z are negative, the candidate is infeasible, go to step 3.

Step 3: return to the feasible cone

Move from the current feasible point x toward z :

$$x \leftarrow x + t(z - x),$$

with the largest feasible step

$$t^* = \min_{i: z[i] < 0} \frac{x[i]}{x[i] - z[i]}.$$

- At least (an often exactly) one coefficient reaches zero, so the support is reduced.
- The cost decreases along the segment (x, z) by convexity!
 - This guarantees that the same support is never visited twice, since all operators in AS reduce the cost, and a given support leads to a fixed residual error.

Then go back to step 2.

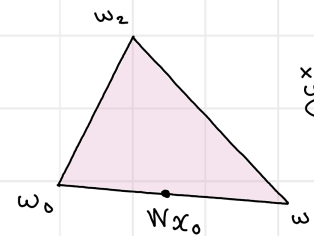
Summary of Active-Set

Problem: $\min_{x \geq 0} \left\| \underset{\substack{\in \mathbb{R}^3 \\ \text{normalized}}}{y} - \left[\underset{\substack{\text{normalized}}}{w_0, w_1, w_2} \right] x \right\|_2^2$

Initialization: $S_0 = \{0, 1\}$, $x_0 = [\ast, \ast, 0]$

First Step: add $\{2\}$ to S_0 ; $S_1 = \{0, 1, 2\}$

Second Step: $z = w^T y$, $wz = y$



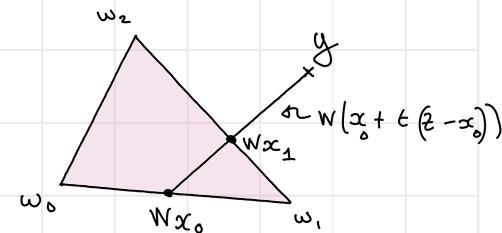
$$\overset{x}{y} = wz$$

z contains negative entries

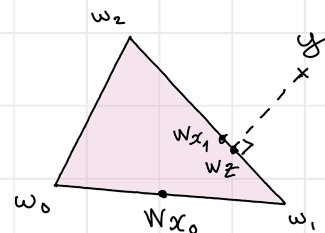
Third Step: $t^* = \frac{x_0[0]}{x_0[0] - z[0]}$

$$x_1 = x_0 + t^*(z - x_0)$$

$$S_2 = \{1, 2\}$$



Second Step (2d iteration): $z = [w_1, w_2]^T y \geq 0$



wz is the orthogonal projection of y on the cone $\langle W \rangle^+$

Active-Set: what is guaranteed?

Every accepted step decreases the NNLS cost.

A support is not revisited unless the optimum has been reached. Therefore AS terminates after finitely many support changes, although the worst-case number of supports is exponential.

AS is especially attractive when:

- the unknown is a single vector
- the solution is sparse
- initialization is close to the solution

HALS for matrix NNLS

Why Active-Set does not scale to batches

Matrix NNLS is

$$\min_{H^T \geq 0} \|Y - WH^T\|_F^2.$$

Each column of H^T is a separate NNLS problem, potentially with a different support.

Solving restricted linear systems independently is expensive.

HALS strategy

Work on rank-one terms, alternatively.

HALS: the matrix view

$$\arg\min_{\substack{U, V \geq 0}} \|F - UV\|_F^2 = \sum \left(\left(\text{row } i \text{ of } F - \text{row } i \text{ of } UV \right)^2 \right)^+$$

HALS is exact alternating optimization:

- each row update decreases the objective
- all right-hand sides are updated together
- with nonzero columns of U , convergence is to the global NNLS minimizer

Lecture 3: we will implement HALS to compute NMF and Nonnegative Tucker decomposition!

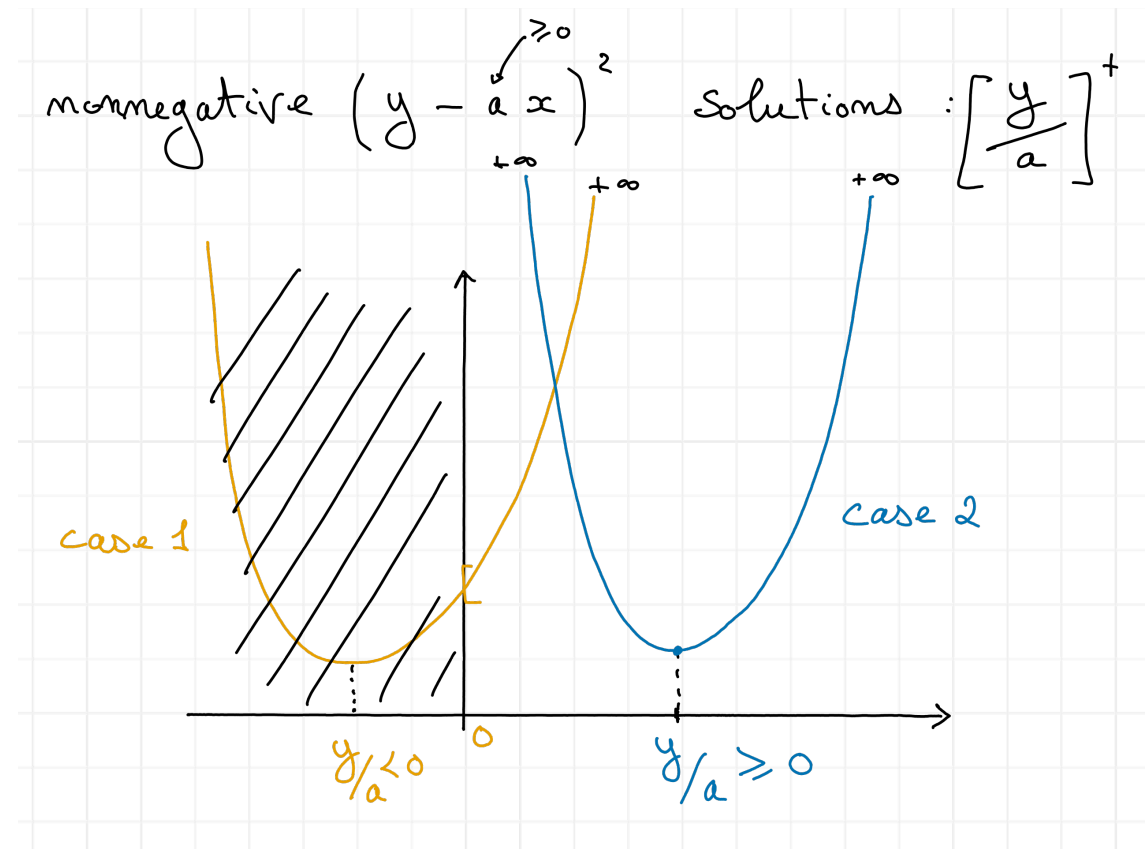
The one-dimensional NNLS building block

For $a \neq 0$ and Z fixed,

$$\arg \min_{h \geq 0} \|Z - ah^T\|_F^2 = \left[\frac{a^T Z}{\|a\|_2^2} \right]_+.$$

Each entry is just a scalar quadratic minimization:

$$h[i] = \max \left(0, \frac{a^T Z[:, i]}{\|a\|_2^2} \right).$$



HALS update

To update row j of H^T , remove its current contribution:

$$Z = Y - U[:, -j] H^T [-j, :].$$

Then update the whole row at once:

$$H^T [j, :] \leftarrow \left[\frac{U[:, j]^T Z}{\|U[:, j]\|_2^2} \right]_+.$$

This is an exact minimization over one block, because the cost is separable with respect to entries in a single row.

Multiplicative Updates

A common update pattern

Multiplicative Updates (MU) preserve nonnegativity by construction:

$$x^{(k+1)} = x^{(k)} * \frac{\text{nonnegative numerator}}{\text{nonnegative denominator}}.$$

For NNLS:

$$x^{(k+1)} = x^{(k)} * \frac{U^T y}{U^T U x^{(k)}}.$$

For NNKL:

$$x^{(k+1)} = x^{(k)} * \frac{U^T (y / U x^{(k)})}{U^T \mathbf{1}}.$$

All products and divisions are elementwise.

MU as a gradient ratio

The gradients can be split into positive and negative parts:

$$\nabla f = \nabla^+ f - \nabla^- f.$$

Then MU has the mnemonic form

$$x^{(k+1)} = x^{(k)} * \frac{\nabla^- f(x^{(k)})}{\nabla^+ f(x^{(k)})}.$$

At a stationary point, the ratio tends to one wherever $x > 0$.

MM: optimize a majorant

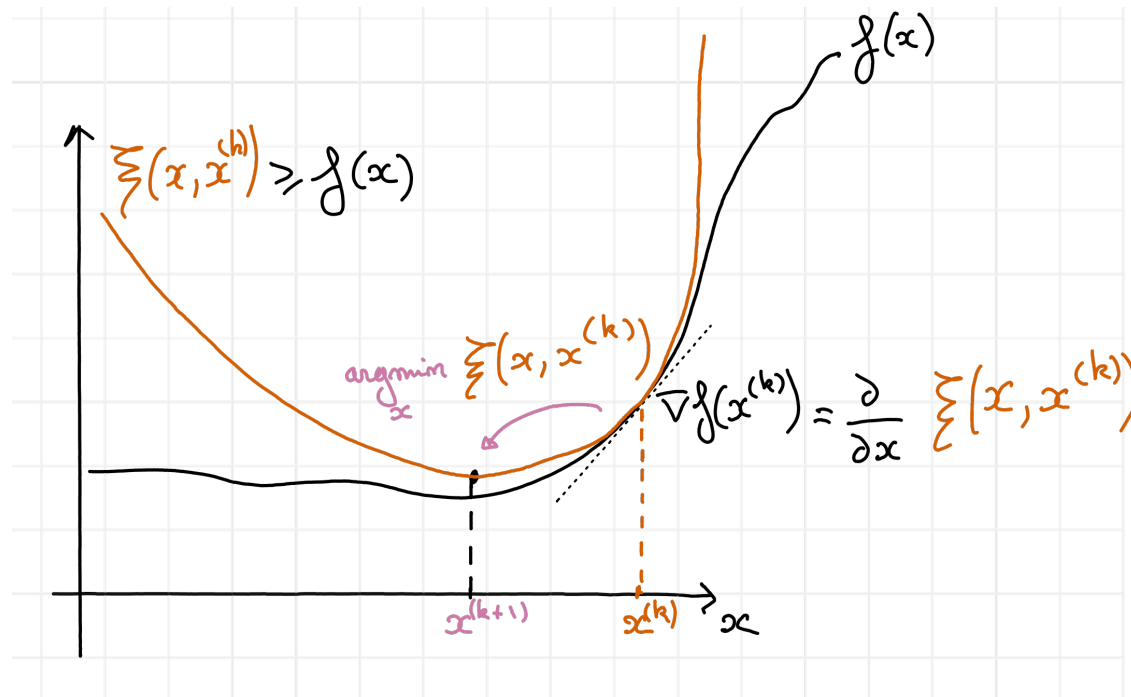
At the current iterate $x^{(k)}$, build a surrogate $\xi(x, x^{(k)})$ such that

$$\xi(x, x^{(k)}) \geq f(y, Ux), \quad \xi(x^{(k)}, x^{(k)}) = f(y, Ux^{(k)}).$$

Then take the next iterate by minimizing the surrogate:

$$x^{(k+1)} \in \arg \min_{x \geq 0} \xi(x, x^{(k)}).$$

The important design goals are **global upper-bounding**, **tightness**, and a tractable minimization.



The majorant touches the cost at $x^{(k)}$ and is easier to minimize. Its minimizer gives a guaranteed descent step.

Jensen constructs the majorant

Jensen's inequality gives

$$f\left(y[i], \sum_j \lambda_{i,j} u_{i,j}\right) \leq \sum_j \lambda_{i,j} f(y[i], u_{i,j}).$$

Convexity turns one coupled loss into a sum of simpler losses.

Let

$$\hat{y}[i] = (Ux^{(k)})[i], \quad \lambda_{i,j} = \frac{U[i,j]x^{(k)}[j]}{\hat{y}[i]}, \quad u_{i,j} = \frac{x[j]}{x^{(k)}[j]} \hat{y}[i].$$

For each observation i ,

$$\sum_j \lambda_{i,j} = 1,$$

so the current reconstruction defines a convex combination.

The Jensen majorant is separable

We get

$$\sum_j \lambda_{i,j} u_{i,j} = \sum_j U[i, j] x[j] = (Ux)[i].$$

Therefore the Jensen bound is the majorant

$$\xi(x, x^{(k)}) = \sum_{i,j} \frac{U[i, j] x^{(k)}[j]}{\hat{y}[i]} f\left(y[i], \frac{x[j]}{x^{(k)}[j]} \hat{y}[i]\right).$$

Each term depends on only one coefficient $x[j]$, so the minimization separates across j .

From the majorant to MU

For every coefficient $x[j]$, set the derivative of the majorant to zero:

$$0 = \sum_i U[i, j] \frac{\partial f}{\partial z} \left(y[i], \frac{x[j]}{x^{(k)}[j]} \hat{y}[i] \right).$$

The two losses give two scalar equations:

Euclidean loss

$$\frac{\partial f}{\partial z} = z - y$$

$$x^{(k+1)} = x^{(k)} * \frac{U^T y}{U^T U x^{(k)}}.$$

KL divergence

$$\frac{\partial f}{\partial z} = 1 - \frac{y}{z}$$

$$x^{(k+1)} = x^{(k)} * \frac{U^T (y / (U x^{(k)}))}{U^T \mathbf{1}}.$$

MM guarantees descent

Because the surrogate is tight at the current iterate,

$$f(x^{(k+1)}) \leq \xi(x^{(k+1)}, x^{(k)}) \leq \xi(x^{(k)}, x^{(k)}) = f(x^{(k)}).$$

MU decreases the objective, but descent alone does not guarantee a fast rate or a unique limit point.

Positivity and zero-locking

If an entry becomes exactly zero, multiplication keeps it zero forever. This is **zero-locking**.

A common safeguard is to enforce $x \geq \epsilon$:

$$x^{(k+1)} \leftarrow \max \left(x^{(k)} * \frac{\nabla^- f}{\nabla^+ f}, \epsilon \right).$$

Strict positivity also avoids division by zero and helps with the KL curvature near zero.

Why MU is natural for NNKL

For Poisson data,

$$Y[i] \mid x \sim \mathcal{P}((Ux)[i]).$$

The latent contribution of component j to observation i is proportional to

$$\frac{U[i, j]x[j]}{(Ux)[i]}.$$

The EM algorithm updates these latent responsibilities, then re-estimates x . The resulting update is exactly NNKL MU.

Rank-one approximations

Rank-one KL approximation

For nonnegative Y , consider

$$\min_{w, h \geq 0} D_{\text{KL}}(Y, wh^T).$$

The optimal scaling conditions imply

$$h^*[j] \sum_i w^*[i] = \sum_i Y[i, j],$$

and

$$w^*[i] \sum_j h^*[j] = \sum_j Y[i, j].$$

Thus the rank-one factors are proportional to the column and row marginals of Y .

Rank-one Frobenius approximation

For nonnegative Y ,

$$\min_{w, h \geq 0} \|Y - wh^T\|_F^2$$

is solved by the leading rank-one SVD approximation, made nonnegative by taking absolute values of the leading singular vectors.

The Perron-Frobenius theorem guarantees nonnegative leading singular vectors for positive matrices.

What to remember

NNLS

- cone projection
- KKT support conditions
- Active-Set for sparse vectors
- HALS for matrix updates

NNKL

- Poisson likelihood
- difficult curvature near zero
- optimal scaling
- MU and EM

The optimization core of nonnegative LRA

Lecture 1 gave the model:

$$\min_{W, H \geq 0} \|Y - WH^T\|_F^2.$$

Lecture 2 opened the inner loop:

$$\min_{x \geq 0} f(y, Wx).$$

The same regression problems reappear inside NMF, nonnegative CP, nonnegative Tucker, and many other structured models.

Any questions ?

Important

Link for Lecture 3 lab session: <https://github.com/cohenjer/HDR/tree/main/cours/Lecture%203>